



INTELLIGENT INTRUSION DETECTION SYSTEM: DETECTING CYBER THREATS WITH MACHINE LEARNING



Introduction

To develop an efficient machine-learning-based intrusion detection system that accurately identifies network threats in real time, while providing users with an interactive FastAPI-React interface for seamless prediction, analysis, and visualization.

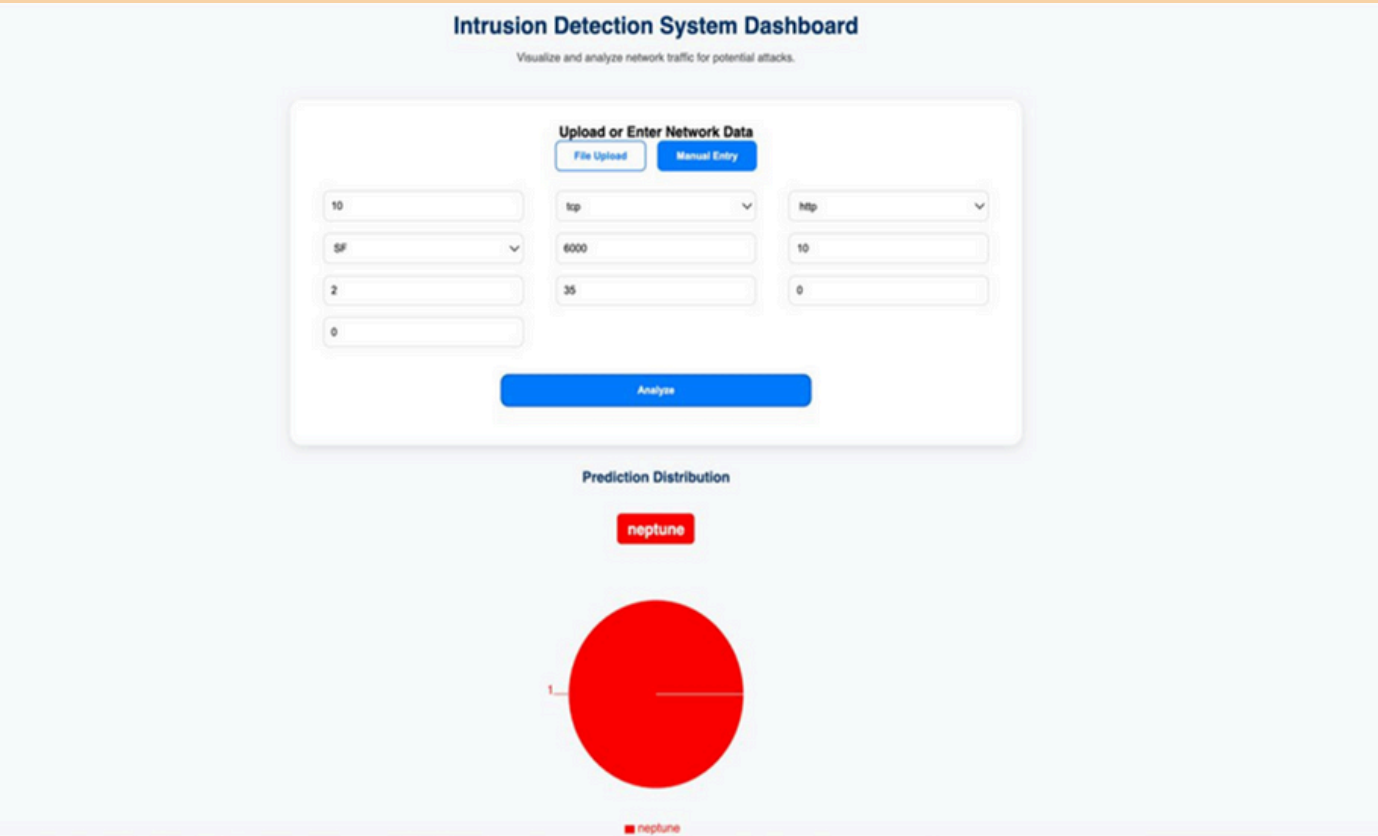
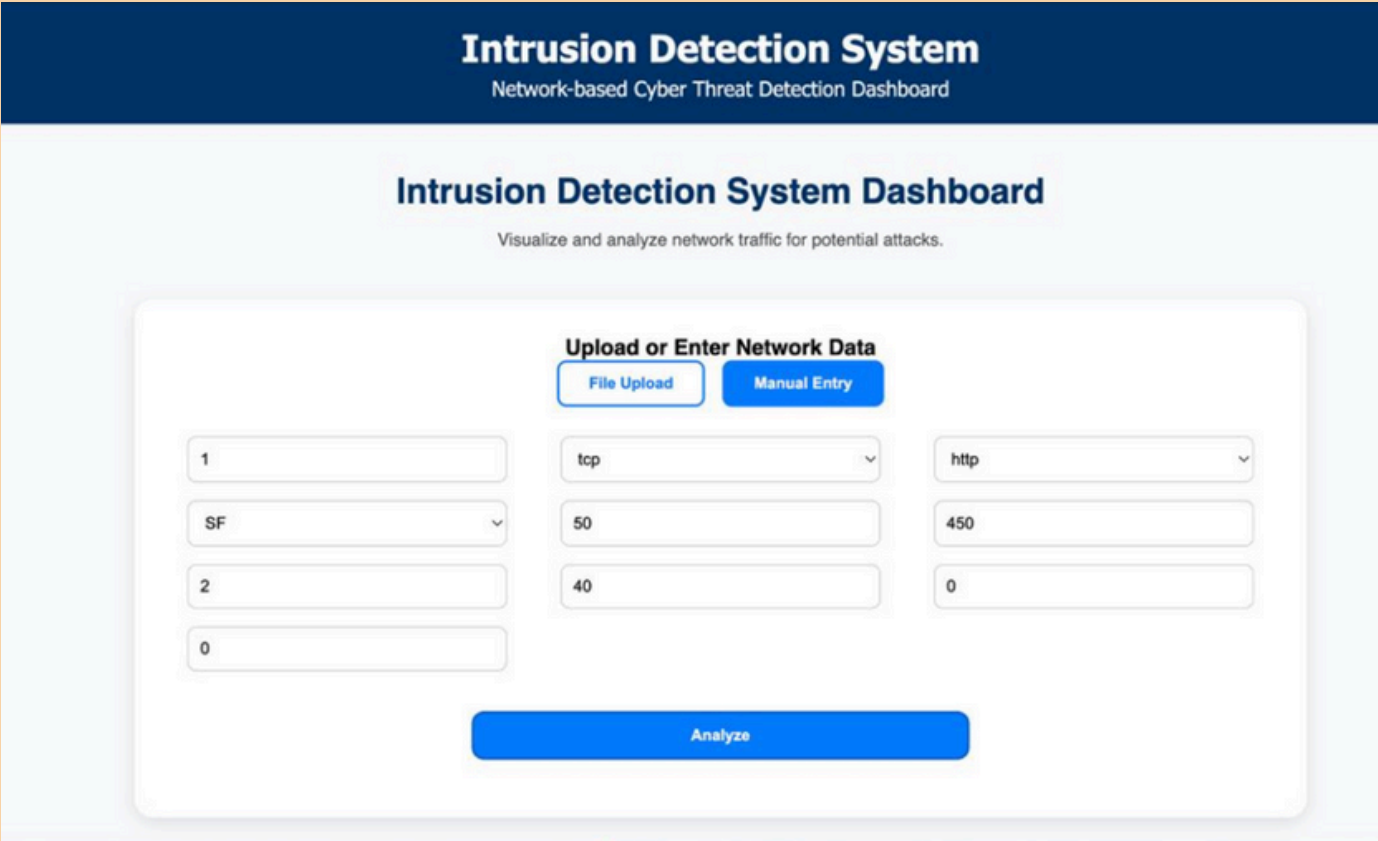
ABSTRACT

This project presents a machine learning-based intrusion detection system that analyzes network traffic using the NSL-KDD dataset. An XGBoost model identifies normal and intrusive activity, while a FastAPI-React interface enables real-time predictions with improved detection accuracy and system reliability.

System Requirements

- 64-bit system with Intel/AMD processor
- 4–8GB RAM for stable model execution
- At least 20GB free storage for datasets and logs
- Python environment with: Scikit-learn, Pandas, NumPy, XGBoost, FastAPI
- Node.js & React for building the frontend
- Visual Studio Code for development
- NSL-KDD dataset for training and testing

Results & Analysis



Conclusion

This project addresses modern cybersecurity challenges by delivering an efficient ML-driven intrusion detection system that streamlines threat identification through automated analysis, real-time predictions, and an intuitive user interface.

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